



DATABRICKS FUNDAMENTALS

Architecture, Workspace, Clusters & Notebooks

A hands-on setup session — from platform concepts to your first running notebook

Duration: 60 minutes

Format: Concept + Live Hands-On

Session Agenda

Six segments across 60 minutes

1	Databricks Architecture Overview	10 min
2	Workspace Tour	8 min
3	Clusters — Concepts	10 min
4	Hands-On: Create a Cluster	10 min
5	Notebooks — Concepts + Hands-On	15 min
6	Wrap-Up + Q&A	7 min

Why Start With Architecture?



Security boundary

Understand what lives in your cloud account vs. what Databricks manages for you.



Cost visibility

Compute runs in your cloud account — architecture explains where your spend goes.



Everything else builds on this

Workspaces, clusters, and notebooks all sit inside this two-plane model.

Control Plane vs. Data Plane

The core architectural split behind every Databricks deployment



Control Plane

Managed by Databricks

- Web UI & Workspace
- Notebook source & version history
- Jobs scheduler
- Cluster manager (orchestration)



Data Plane

Lives in your cloud account (AWS / Azure / GCP)

- Clusters (driver + worker VMs)
- Your data in cloud storage
- Network & security groups
- Actual query & job execution

Workspace Tour

What you'll see the moment you log in



Workspace

Notebooks, folders, personal & shared spaces



Catalog

Databases, tables, and Unity Catalog volumes



Workflows

Scheduled jobs and multi-task pipelines



Compute

Create & manage clusters and SQL warehouses



SQL

Query editor and dashboards

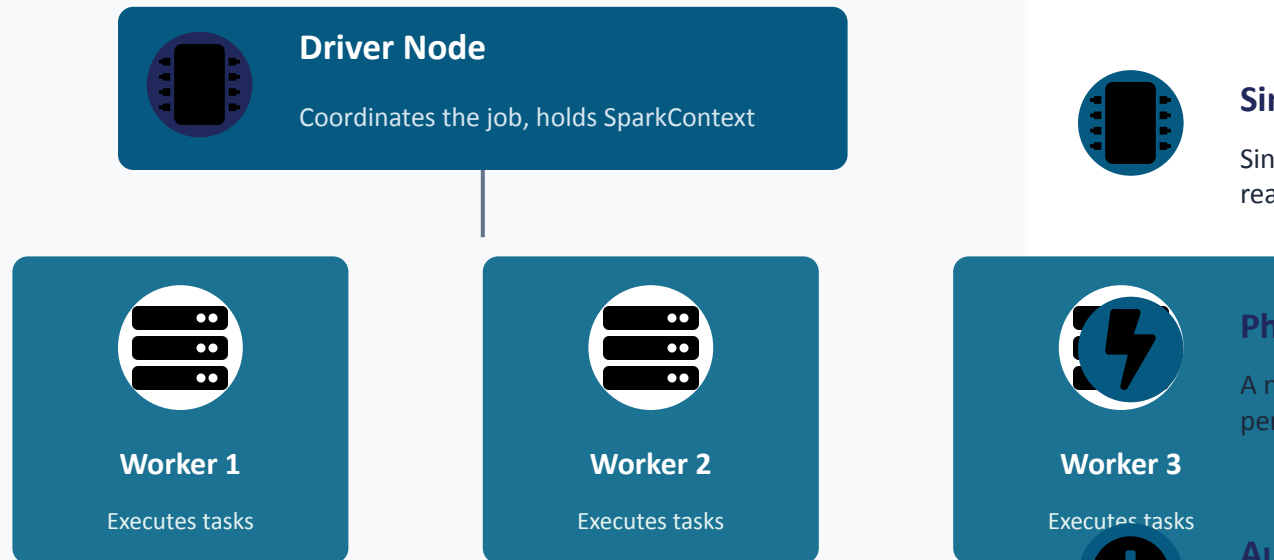


Workspace folders vs. Repos

Workspace folders hold notebooks directly (personal or shared); Repos connect a folder to a Git repository for version-controlled, collaborative development.

Clusters — Core Concepts

Anatomy of a Cluster

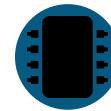


Autoscaling adjusts worker count between your min/max settings



All-Purpose vs. Job Clusters

All-purpose: shared, interactive dev work. Job clusters: spun up per job, terminate on completion — cheaper for scheduled runs.



Single-Node vs. Multi-Node

Single-node = driver only, great for cost-conscious dev/test. Multi-node = driver + workers for real parallel processing.



Photon Engine

A native vectorized query engine that accelerates SQL and DataFrame workloads — enable it per cluster.



Auto-Termination

Shuts an idle cluster down automatically — the single biggest cost-control lever in Databricks.

Cluster Configuration Choices

What you'll set when creating a cluster

Databricks Runtime version

The managed Spark + libraries bundle — pick LTS versions for stability.

Node type

VM size for driver/workers — balance memory, cores, and cost.

Autoscaling min/max

Cluster grows and shrinks worker count within these bounds as load changes.

Auto-termination

Minutes of inactivity before the cluster shuts down — always set this in dev/training.

Cluster policy

Org-defined guardrails (max size, allowed instance types) applied by admins.



HANDS-ON

Create Your First Cluster

1

Go to Compute → Create Cluster

Give it a clear name (e.g. yourname-training)

2

Choose Single Node

Keeps cost near-zero for this exercise

3

Set autoscaling / worker count

Leave defaults if using single node

4

Set auto-termination to 30-45 min

This is the most important cost-safety step

5

Click Create Cluster and wait

Status turns green (Running) in 3-5 minutes

Notebooks — Core Concepts



Magic Commands

%python

Run a Python cell (default language)

%sql

Run SQL directly against tables

%md

Render Markdown for documentation

%sh

Run shell commands on the driver node



dbutils basics

dbutils.fs.ls() browses storage paths; dbutils.widgets creates interactive input parameters.



Cell execution & state

Cells share one session state — run order matters, not just top-to-bottom position.



Version history & Git

Every notebook keeps revision history; link to Repos for full Git-based collaboration.



Attach to a cluster

A notebook needs a running cluster attached before any cell will execute.



HANDS-ON

Build Your First Notebook

1

Create a notebook

Workspace → Create → Notebook, attach to your cluster

2

Add a %md title cell

Document the notebook's purpose in Markdown

3

Load a sample dataset

%python — read a built-in sample table into a DataFrame

4

Query it with %sql

Switch language mid-notebook to explore the same data

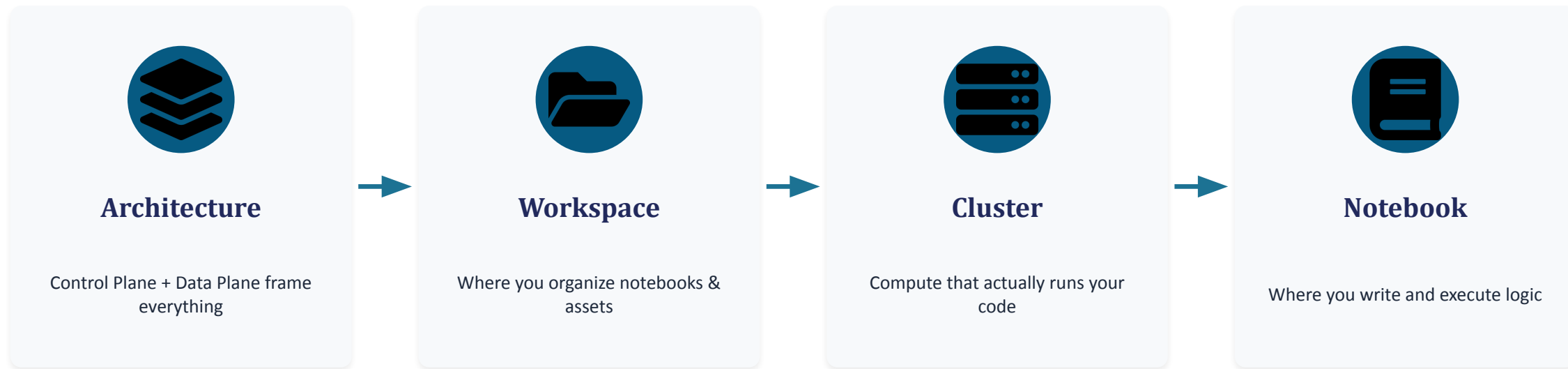
5

Check version history

Open the clock icon to see the notebook's revision history

How It All Connects

Architecture → Workspace → Cluster → Notebook



What's Next

Where this session leads in future modules



Unity Catalog

Centralized governance for data, tables, and permissions across workspaces.



Delta Lake Basics

ACID transactions, time travel, and reliable tables on top of cloud storage.



DBFS & Volumes

Managing files and unstructured data alongside your tables.



Questions?

Thank you for joining — let's get hands-on next time with Delta Lake